

<page_start no = 1>

<sol_start id=1>

1. Astigmatism

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<sol_start id=2>

2. Repulsion

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<sol_start id=3>

3. Laterally inverted

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<sol_start id=4>

4. Babr the image of fish.

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<sol_start id=5>

5. The focal length of the lens = $\frac{1}{1.5} \text{m} \left(\text{as } p = \frac{1}{f} \text{ or } r = \frac{1}{p} \right)$

$$\text{So, } \frac{1}{1.5} \times \frac{10}{10} = \frac{\cancel{10}}{\cancel{15}} = \frac{2}{3} \text{m} = 0.67 \text{m}$$

The lens is converging or convex lens.

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6. We know $I = \frac{q}{t}$

Given, $I = 5A$, $t = 30$ minutes

$$q = I \times t$$

$$= 5 \times 30 \times 60 \text{ c}$$

$$= 150 \times 60 \text{ c}$$

$$= 9000 \text{ c}$$

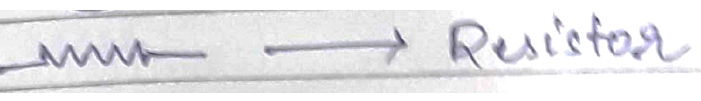
<sol_end>

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7. Circuit symbol of five important components are -

(1)  $\rightarrow$ Cell

(2)  $\rightarrow$ Bulb

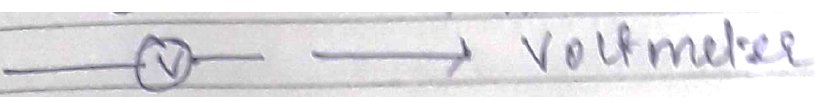
(3)  $\rightarrow$ Resistor

(4)  $\rightarrow$ Key

(5)  $\rightarrow$ Switch (OFF)

(6)  $\rightarrow$ Switch (ON)

(7)  $\rightarrow$ Ammeter

(8)  $\rightarrow$ Voltmeter

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8. Given : Lens nature is convex

Focal length = +12 cm

$h_o = 5$ cm

Object distance = -20 cm

Using lens, formula

$$\frac{1}{v} = \frac{1}{u} + \frac{1}{f}$$

$$\text{Or, } \frac{1}{v} - \frac{1}{-20} = \frac{1}{12}$$

$$\text{Or } \frac{1}{v} + \frac{1}{20} = \frac{1}{12}$$

$$\text{Or } \frac{1}{v} = \frac{1}{12} - \frac{1}{20}$$

$$\text{Or, } \frac{1}{v} = \frac{5-3}{60} = \frac{2}{60} \text{ cm}$$

$$\text{Or } v = \frac{60}{2} \text{ cm} = 30 \text{ cm} \rightarrow \text{image distance}$$

$$\text{Size of image} \rightarrow \frac{v}{u} = m (\text{magnification})$$

$$\text{Or, } \frac{30}{-20} = m$$

$$\text{Or } \frac{-3}{2}$$

$$\frac{-3}{2}, \frac{h_i}{h_o} \Rightarrow \frac{-3}{2}, \frac{h_i}{5\text{cm}} \Rightarrow -15\text{cm} = 2h_i$$

$$\text{Or } h_i = \frac{-15}{2} = -7.5\text{cm}$$

Hence, the nature of image is real, inverted and magnified.

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9. The charge is to be moved from higher potential to lower potential, i.e. $10\text{v} \rightarrow 5\text{v}$.

$$V_B - V_A = \frac{W_{AB}}{q_o}$$

$$\text{Or } 10\text{v} - 5\text{v} = \frac{5\text{j}}{q_o}$$

$$\text{Or, } 5\text{v} = \frac{5\text{j}}{q_o}$$

$$\text{Or } 5q_o = 5\text{j}$$

$$\text{Or, } q_o = \frac{5\text{j}}{5\text{v}} = 1\text{c}$$

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10. PART-I

Pole : it is a geometrical centre of the reflecting surface. It is represented by letter P.

Radius of curvature : The radius of the sphere of which the mirror is a part of.

Principal Axis : The line joining the pole and centre of curvature of the mirror.

Principal focus : it is a point on the principal axis at which rays parallel to principal axis meet after reflection.

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PART-II

The range of distance is in between the pole and the focus. i.e 0 cm to 20 cm.

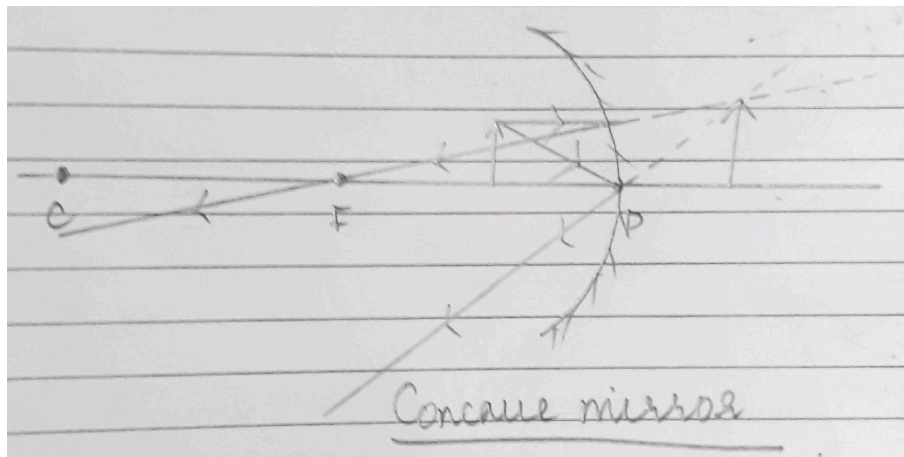
Characteristics of observed image

(i) Magnified image

(ii) Formed behind the mirror.

(iii) Virtual image

RAY DIAGRAM



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- 11.(a) The sun appears reddish in the early morning as it is closer to the horizon and undergoes atmosphere it refraction.
- (b) Power of accommodation of human eye means the ciliary muscle would focus the image on retina when the image is far or nearer.
- (c) The least distance of distinct vision of human eye is 25 cm.

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